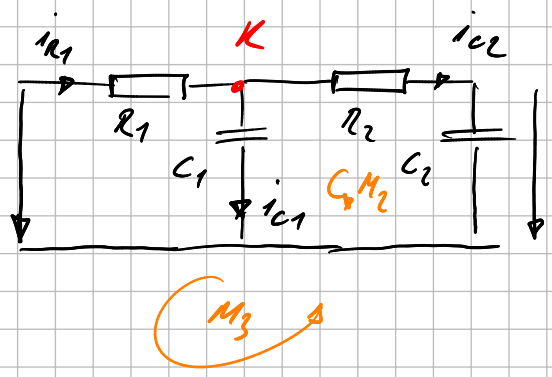
Ersatzschaltbild

System Eingang $u = u_E$



$u_{C2} = y$ System Ausgang

1 DGL

$$\boxed{\begin{aligned} u_C &= \frac{1}{C}\int i_C\,dt \\ i_C &= C\dot{u}_C \\ u_R &= R\,i \end{aligned}}$$

K: $\quad i_{R1} = i_{C1} + i_{C2}$

M_2: $\quad u_{C1} = u_{R2} + u_{C2}$

M_3: $\quad u_E = u_{R1} + u_{R2} + u_{C2}$

$$u_E = \underbrace{R_1(i_{C1} + i_{C2})}_{u_{R1}} + \underbrace{R_2 i_{C2}}_{u_{R2}} + u_{C2}$$

$$= R_1(\underbrace{C_1\dot{u}_{C1}}_{i_{C1}} + \underbrace{C_2\dot{u}_{C2}}_{i_{C2}}) + R_2\underbrace{C_2\dot{u}_{C2}}_{i_{C2}} + u_{C2}$$

$$= R_1(C_1\underbrace{\dot{(u_{R2} + u_{C2})}}_{\dot{u}_{C1}} + C_2\dot{u}_{C2}) + R_2C_2\dot{u}_{C2} + u_{C2}$$

$$= R_1(C_1(\underbrace{R_2\dot{i}_{C2}}_{\dot{u}_{R2}} + \dot{u}_{C2}) + C_2\dot{u}_{C2}) + R_2C_2\dot{u}_{C2} + u_{C2}$$

$$= R_1(C_1(R_2\underbrace{C_2\ddot{u}_{C2}}_{\dot{i}_{C2}} + \dot{u}_{C2}) + C_2\dot{u}_{C2}) + R_2C_2\dot{u}_{C2} + u_{C2}$$

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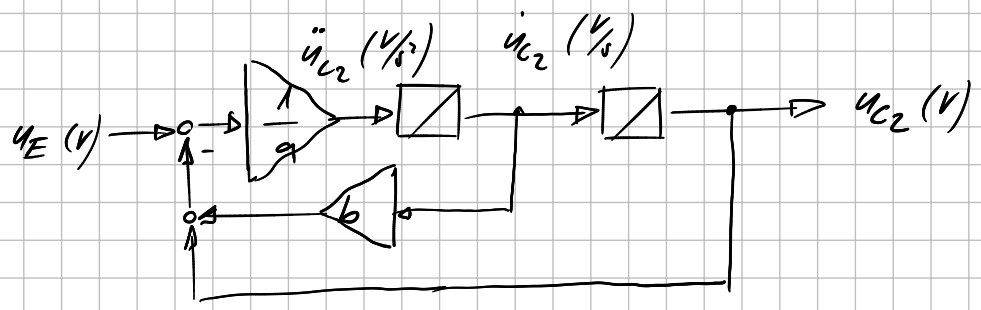
$$\rightarrow u_E = \underbrace{R_1C_1R_2C_2}_{a = R_1C_1R_2C_2}\ddot{u}_{C2} + \underbrace{(R_1C_1 + R_1C_2 + R_2C_2)}_{b = R_1C_1 + R_1C_2 + R_2C_2}\dot{u}_{C2} + u_{C2} \quad //$$

$a = 4{,}87 \cdot 10^{-6}$ $\qquad$ $b = 0{,}0066$

3 Blockschaltbild

$$a\ddot{u}_{C_2} + b\dot{u}_{C_2} + u_{C_2} = u_E \rightarrow \ddot{u}_{C_2} = \frac{1}{a}\left(u_E - \left(b\dot{u}_{C_2} + u_{C_2}\right)\right)$$

2 Ableitungen → System 2ter Ordnung → 2 Integratoren



4 Übertragungsfunktion ↔ Frequenzgang

$$a\ddot{u}_{C_2} + b\dot{u}_{C_2} + u_{C_2} = u_E$$

$$as^2 U_{C_2}(s) + bs\,U_{C_2}(s) + U_{C_2}(s) = U_E(s)$$

$$(as^2 + bs + 1)\,U_{C_2}(s) = U_E(s)$$

$$U_{C_2}(s) = \underbrace{\frac{1}{as^2+bs+1}}_{G(s)}\,U_E(s) \qquad G(s) = \frac{1}{as^2+bs+1} = \frac{U_{C_2}(s)}{U_E(s)} \quad //$$

$$\boxed{G(j\omega) = G(s)\Big|_{s=j\omega}} = \frac{1}{a(j\omega)^2 + bj\omega + 1} = \frac{1}{1 - a\omega^2 + jb\omega} \quad //$$

Statische Verstärkung

Anhand DGL

$$u_E = a\ddot{u}_{C_2} + b\dot{u}_{C_2} + u_{C_2}$$

statisch → Ableitungen verschwinden

$$u_E = \underbrace{1}_{K} \cdot u_{C_2} \quad //$$

für $t \to \infty$

$$u_E \to 1 \to u_{C_2}$$

Einfacher anhand Übertragungsfunktion

$$G(s)\Big|_{s=0} = 1 \frac{V}{V} \quad //$$

$$G(s) = \frac{B(s)}{A(s)} \qquad B(s) = 1 \qquad A(s) = as^2 + bs + 1$$

Nullstellen

$$B(s) = 1 \overset{!}{=} 0$$

$$\to z = \{\} \quad //$$ (keine Nullstellen)

Polstellen

$$A(s) = as^2 + bs + 1 \overset{!}{=} 0$$

$$\to p_{1,2} = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4a}}{2a} = -172{,}9 \,;\, -1185{,}2 \quad //$$

(zwei reelle Polstellen)

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$$T_1 = \frac{1}{-p_1} = 5{,}78 \text{ msec} \qquad T_2 = \frac{1}{-p_2} = 0{,}84 \text{ msec}$$

Polynomform

$$G(s) = \frac{1}{as^2+bs+1} = \frac{\overbrace{1/a}^{b_0}}{s^2+\underbrace{b/a}_{a_1}s+\underbrace{1/a}_{a_0}} = \frac{2{,}05\cdot 10^5}{s^2+1358s+2{,}05\cdot 10^5}$$

Zeitkonstanten Form

$$G(s) = \frac{1}{(T_1 s+1)(T_2 s+1)} = \frac{1}{(5{,}78\cdot 10^{-3}s+1)(0{,}84\cdot 10^{-3}s+1)}$$

Pol – Nullstellen Form

$$G(s) = \frac{p_1 p_2}{(s-p_1)(s-p_2)} = \frac{2{,}05\cdot 10^5}{(s+1185)(s+173)}$$

$$\left(G(s) = \frac{(-p_1)\cdot(-p_2)}{(s-p_1)(s-p_2)} \right)$$

Achtung:
Für ein System 3ter Ordnung gilt

$$G(s) = \frac{-p_1 p_2 p_3 K}{(s-p_1)(s-p_2)(s-p_3)} = \frac{(-p_1)\cdot(-p_2)\cdot(-p_3)K}{(s-p_1)(s-p_2)(s-p_3)}$$

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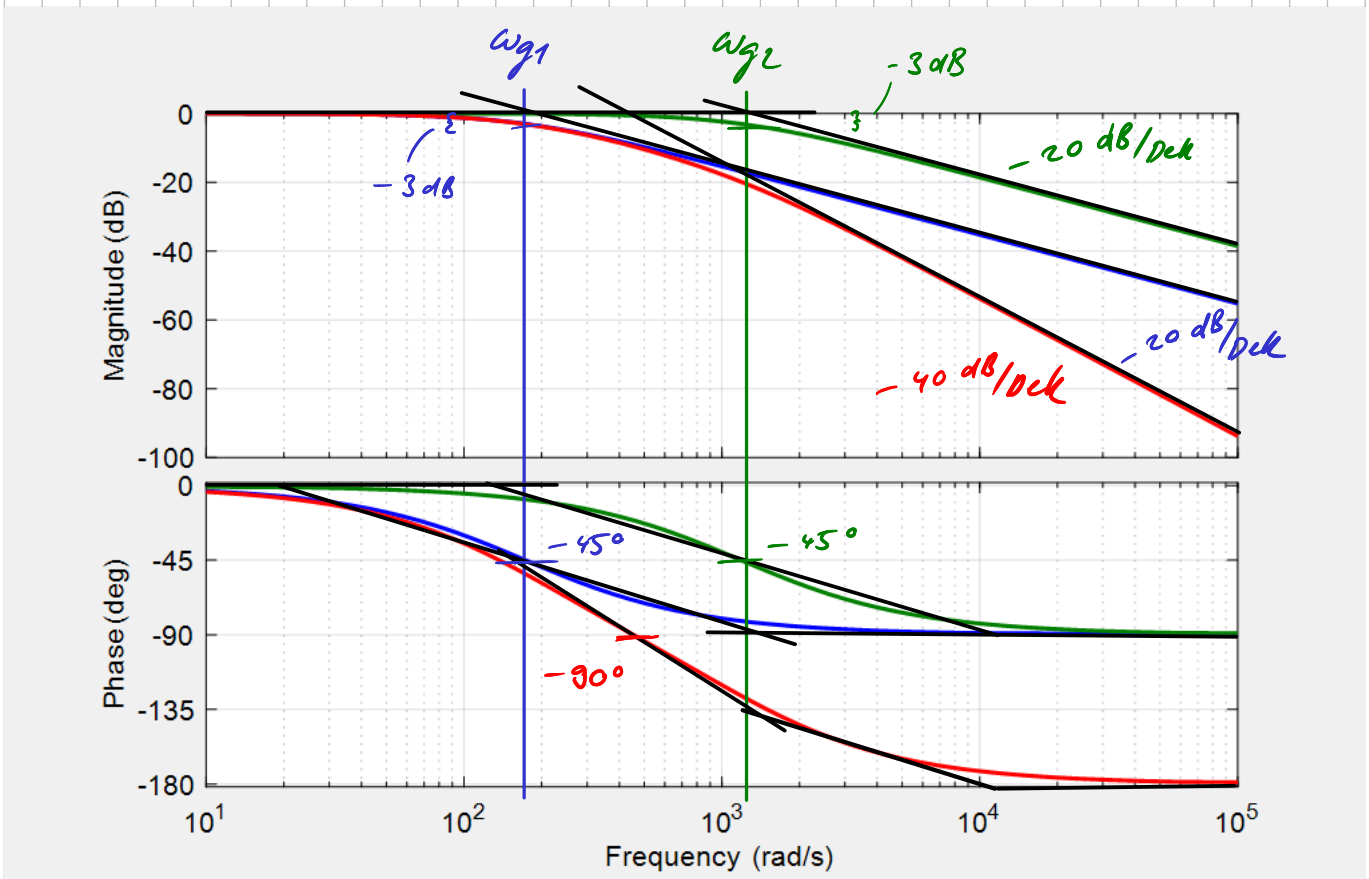
$$PT_2: \quad G(s) = \underbrace{\frac{172{,}9}{s+172{,}9}}_{PT_1:\ G_1(s)} \cdot \underbrace{\frac{1185{,}2}{s+1185{,}2}}_{PT_1:\ G_2(s)} \qquad G(0)=1$$

$\omega_{g1} = 172{,}9\ \mathrm{rad/sec}$ $\qquad$ $\omega_{g2} = 1185{,}2\ \mathrm{rad/sec}$

$G_1(0) = 1$ $\qquad$ $G_2(0) = 1$

Amplitudengang: $dB(G(0)) = dB(G_1(0)) + dB(G_2(0))$ $\qquad$ $G_1(\omega_{g1}) = 0\,dB\ (=1)$

Phasengang: $\angle(G(0)) = \angle(G_1(0)) + \angle(G_2(0))$ $\qquad$ $G_2(\omega_{g1}) = 0\,dB\ (=1)$